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The Mental Health Burden of Endometriosis: A Narrative Review

Ana Clara Pimenta Servidoni, Hanna Surmann, MD, Michel Canis, MD PhD, Hugh S. Taylor, MD, Jose Alexandre Crippa, MD PhD, Mauricio S. Abrao, MD PhD

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Authors and Affiliations: Ana Clara Pimenta Servidoni 1,2, Hanna Surmann MD 1,3, Michel Canis MD PhD 4, Hugh S. Taylor MD 1, Jose Alexandre Crippa MD PhD 5,6, Mauricio S. Abrao MD PhD 7,8

1. Department of Obstetrics, Gynecology and Reproductive Sciences, Yale School of Medicine, New Haven, Connecticut, USA
2. Faculdade Santa Marcelina, São Paulo, Brazil
3. Department of Gynecology and Obstetrics, University of Muenster, Albert-Schweitzer-Campus 1, 48149 Münster, Germany
4. Department of Gynecological Surgery, Clermont-Ferrand, France
5. Department of Neuroscience and Behavioral Sciences, Ribeirão Preto Medical School, University of São Paulo, Brazil.
6. Psychiatry Department, Faculdade de Medicina da Universidade de São Paulo, São Paulo, SP, Brazil
7. Gynecology Division, Beneficência Portuguesa de São Paulo, São Paulo, SP, Brazil
8. Gynecology Division, Obstetrics and Gynecology Department, Faculdade de Medicina da Universidade de São Paulo, São Paulo, SP, Brazil

Corresponding Author: Mauricio Simões Abrão Gynecology Division, Beneficência Portuguesa de São Paulo Rua Colômbia, 332, São Paulo, SP, Brazil Phone: +551131503344 Email: msabrao@mac.com

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Abstract

This narrative review synthesizes existing literature to elucidate the complex interplay between endometriosis and mental health disorders, exploring their prevalence, underlying mechanisms, and the impact of systemic failures and misinformation. Compared with women without endometriosis, patients with endometriosis have significantly higher rates of mental health disorders, including a 1.5-times higher risk of depression, a 1.4-times higher risk of anxiety, and elevated rates of body dysmorphic disorder and post-traumatic stress disorder. These comorbidities stem from a complex bidirectional relationship involving chronic inflammation, pain sensitization, early-life trauma, and the profound psychosocial impact of infertility. Prolonged diagnostic delays and feelings of invalidation significantly contribute to psychological injury, while social media misinformation heightens distress and drives patients towards unproven interventions. Endometriosis imposes a substantial mental health burden, exacerbated by these systemic issues and misleading narratives. Effective management necessitates integrated, multidisciplinary care, including robust mental health support, and proactive governmental policies to promote specialized centers, combat misinformation, and implement diagnostic advancements, thereby improving patient outcomes and quality of life.

Keywords: endometriosis, mental health, depression, anxiety, diagnostic delay

Essential Points

- Endometriosis imposes a substantial mental health burden, with patients facing heightened risks of depression, anxiety, body dysmorphic disorder, and post-traumatic stress disorder.
- The psychological impact is amplified by systemic failures in care, including prolonged diagnostic delays and medical invalidation, alongside misinformation and fear-driven narratives.

- Complex bidirectional relationships involving chronic inflammation, pain sensitization, early-life trauma, and infertility contribute to the observed psychiatric comorbidities.
- Effective management requires integrated, multidisciplinary care incorporating mental health professionals, supported by proactive governmental policies to establish specialized centers and combat misinformation.

1. Introduction

Endometriosis is defined as the presence of ectopic endometrial tissue outside the uterine cavity and it affects up to 10% of reproductive age women (1). This condition is a major cause of pelvic pain and infertility (2). Dysmenorrhea, dyspareunia, dysuria, dyschezia and chronic pelvic pain are the most common symptoms that can cause both physical and emotional distress (3,4).

The quality of life for women with endometriosis is frequently compromised across multiple dimensions. The pervasive nature of the disease impacts essential aspects of daily living, including school attendance, work productivity, general daily activities, engagement in physical exercise, and the ability to achieve restful sleep (3,5–7). This suffering typically extends beyond the physical lesions, revealing a critical failure in delivering holistic care (8,9). The internal and often invisible nature of the condition exacerbates stigma, rendering women's pain largely unacknowledged and contributing significantly to protracted diagnostic delays (10). Large population-based cohort studies and meta-analyses have demonstrated ~~The~~ that prolonged and unaddressed pain creates a direct pathway to severe mental health disorders, fundamentally impairing the quality of life even further (11–13).

This review synthesizes the impact of endometriosis on mental health disorders, distinguishing itself by analyzing under-recognized comorbidities, introducing the novel 'market of fear' construct, and reconceptualizing diagnostic delays as a systemic diagnostic failure inflicting profound psychological burden. Ultimately, it advocates for comprehensive integrated care and policy reforms, aiming to establish a more nuanced, comprehensive, and patient-centered framework for holistic endometriosis management

2. Methods

This narrative review aimed to summarize the current evidence on the relationship between endometriosis and mental health outcomes. A comprehensive literature search was performed between 2000 and 2026, with a particular emphasis on publications from the last 5-10 years to capture the most recent advancements and build upon existing reviews, primarily using the PubMed and Scopus databases. A particular emphasis was placed on publications from the last 10 years to capture recent advancements and build upon existing reviews. The search combined Medical Subject Headings (MeSH) terms and free-text keywords relating to endometriosis and mental health. The search terms included, but were not limited to, the following: 'endometriosis', 'chronic pelvic pain', 'mental health', 'depression', 'anxiety', 'post-traumatic stress disorder', 'body image', 'quality of life', 'infertility', 'psychological distress', 'inflammation', and 'pain sensitization'.

Findings were synthesized qualitatively. The selection process followed a strict hierarchy of evidence, prioritizing systematic reviews and meta-analyses, followed by large-scale population-based cohort studies. High-quality narrative reviews and observational studies were also considered, followed by animal studies and case reports. The review focused on original research articles published in English. The literature was organized thematically, integrating the biological, psychological and social dimensions of the mental health burden associated with endometriosis.

3. Pathogenesis, pain and inflammation

The pathophysiology of endometriosis is complex and multifactorial and not yet completely understood (14), with current theories encompassing retrograde menstruation (15,16), immune dysfunction characterized by a sustained pro-inflammatory microenvironment (14,17), stem/progenitor cell involvement (17–19), and lymphatic or hematogenous dissemination (20–23). The dynamic interplay among these mechanisms is likely central to disease initiation and progression, and the correlation between age and severity remains inconclusive (24,25).

Similarly, the precise mechanisms underlying endometriosis-associated pain are still being elucidated, involving a persistent inflammatory milieu, peripheral and central sensitization, and aberrant neurogenesis (26). Patients exhibit a substantial immune response, with altered immune cell composition and activation and increased inflammatory cytokine production (IL-6, IL-1 β , TNF- α) (26,27). This inflammatory environment, coupled with elevated nerve growth factor levels and oxidative stress, contributes to sensory nerve activation and hyperexcitability (28–30).

The disease also leads to increased nerve fiber density within and surrounding endometriotic lesions, through enhanced neuroangiogenesis driven by inflammatory cytokines and growth factors (30–32). This results in peripheral sensitization, characterized by increased pain perception, lowered pain thresholds, and cross-sensitization among pelvic organs (30,33). Moreover, central sensitization develops, involving increased excitability of spinal cord neurons and altered gene expression in pain-related brain regions (30,34). In mouse models, endometriosis has been shown to induce pain sensitization as well as anxiety- and depression-like behaviors by modulating brain gene expression and electrophysiology; these central nervous system alterations may underlie both pain sensitization and mood disorders reported in patients (35).

A bidirectional relationship between chronic pain and psychiatric disorders is present in endometriosis, with inflammation being one of the main shared mechanisms. The inflammatory environment in endometriosis patients provides a permissive environment for neuroendocrine and neurotransmitter dysregulation (26). Cytokines may cross the blood–brain barrier and impact brain function by altering neurotransmitter metabolism, neuroendocrine regulation, and neuroplasticity. For example, brain-derived neurotrophic factor (BDNF), which is important for both depression and pain regulation, has been found to be significantly elevated in women with endometriosis as well as in patients with other pain conditions (26,36).

Dysregulation of the hypothalamic–pituitary–adrenal axis may also occur under the influence of inflammatory cytokines, leading to glucocorticoid resistance and altered stress responsivity (26). Indeed, inflammatory markers and BDNF levels differ significantly in depressed or anxious patients who also have endometriosis, and these biochemical changes are significantly associated with pain intensity (26).

Overall, the mechanisms that cause pain and mental health disorders substantially overlap and may be related to the pathogenesis of endometriosis. Psychological stress further

exacerbates this process; mice exposed to prolonged stress showed an increase in endometriotic lesion burden (37). In this respect, endometriosis appears to cause pain and psychiatric disorders through interacting inflammatory, neuroendocrine, and neuroplastic mechanisms, which in turn exacerbate the disease. This bidirectional amplification suggests a self-perpetuating pathophysiological cycle.

4. Endometriosis and mental health disorders

Endometriosis-related pain has a tremendous impact on women's lives, substantially disrupting daily routines and functional capacity. It interferes with educational attainment, career advancement, social functioning, and intimate relationships (38,39). This incessant burden of pain, coupled with persistent daily dysfunction, frequently culminates in chronic fatigue and a pervasive feeling of hopelessness (40). The experience of endometriosis-related pain thus serves as a direct and sustained pathway to mental health deterioration (41). Women with endometriosis exhibit a notably higher prevalence and increased risk of developing depressive disorders, anxiety, and emotional distress compared with the healthy population (41,42), which may also negatively influence treatment adherence and outcomes (43) (Figure 1).

4.1. Depression

Major depressive disorder (MDD) represents a prevalent mental health condition that substantially diminishes quality of life and increases susceptibility to a range of adverse health outcomes. Although there is global consensus regarding the critical need for early detection, clinical guidelines report that fewer than 20% of affected individuals receive adequate treatment (44). Epidemiological data indicate that MDD is approximately twice as common in women as in men (45). Genetics contributes to vulnerability; however, social, environmental, and contextual factors play a central role in increasing the risk of MDD (46), which may partially explain the higher prevalence of depressive symptoms among women with endometriosis compared with the general female population (41,42).

The intrinsic nature of chronic pain associated with endometriosis, characterized by its intensity, persistence, and resistance to conventional treatments, emerges as one of the primary catalysts for the development and exacerbation of depression (41). This continuous physical suffering not only degrades quality of life but also generates a self-perpetuating cycle of fatigue, sleep disturbances, and functional impairment, further exacerbating depressive symptoms (40). Central sensitization plays a key role in this process, as it leads to heightened pain perception beyond the localized pelvic areas. This contributes to nociplastic pain, where even routine physical activities or stimuli trigger disproportionate pain responses that negatively impact mental health and emotional resilience. Women with endometriosis often develop comorbidities like irritable bowel syndrome or migraines, which further intensify sensitization, creating a direct link between physical and psychological suffering (47–50). Neurogenic inflammation and oxidative stress in endometrial lesions exacerbate nociceptor activation, contributing to pain persistence that perpetuates depressive symptoms. Additionally, disordered

sleep caused by pain interference further intensifies psychological stress, creating a feedback loop that undermines recovery (51).

A large cohort study comprising over 70,000 women diagnosed with endometriosis revealed that they were 1.48 times more likely to develop clinically recognized depression and 2.03 times more likely to experience self-directed violence compared with women without the condition (52). Within the endometriosis cohort, pain-related symptoms, prior opioid use, and fatigue emerged as significant risk factors for incident depression (52). These findings are further supported by a recent retrospective investigation of endometriosis-associated chronic pain, which reported a 42.5% prevalence of clinically significant depressive symptoms; depression was also correlated with elevated inflammatory markers, suggesting that more severe inflammatory disease may confer greater vulnerability to depression (26). Consistent with these observations, another study confirmed that chronic pelvic pain and painful defecation significantly worsen depressive symptoms, independently of disease stage (53).

MDD represents a highly impactful psychiatric comorbidity in women diagnosed with endometriosis (54,55). The effects of this disorder extend across all domains of life, compromising functionality, productivity, and the ability to engage in activities that previously provided pleasure and meaning (56). Despite this impact, factors such as social stigma surrounding gynecological conditions and the underdiagnosis of mood disorders in routine clinical settings frequently hinder timely intervention. Chronic depressive symptoms, if left untreated, correlate not only with greater pain intensity and lower adherence to medical recommendations, but also with decreased effectiveness of surgical and pharmacological outcomes, underscoring the urgent need for integrated care approaches (51,57).

Some challenges in diagnosis include brief consultation times, which may restrict opportunities for patients to disclose their symptoms, while misjudging the severity of mood disturbances can lead to missed diagnoses. Additionally, diagnostic accuracy is significantly affected by the clinician's attentiveness to mood-related symptoms and the quality of the therapeutic alliance established during consultations (58).

For optimal management, women with endometriosis and suspected MDD should be promptly referred for specialized mental health evaluation. Effective treatment often involves a combination of pharmacological interventions, such as selective serotonin reuptake inhibitors (SSRIs), and structured, depression-specific psychotherapies, including cognitive behavioral therapy (CBT). These approaches, when integrated, address the complex psychological and physical dimensions of this comorbidity (45).

4.2. Generalized anxiety disorder

Generalized anxiety disorder (GAD) is defined as the presence of excessive and persistent worry and anxiety that are difficult to control (59). Symptoms can include persistent, uncontrollable worry, often accompanied by physical tension, cognitive difficulties, and irritability, causing significant distress or functional impairment for at least six months. This

chronic condition is associated with increased risk of suicidal ideation, cardiovascular morbidity, and greater healthcare utilization (60).

Individuals with endometriosis frequently experience GAD. The persistent and excessive worry about a variety of events or activities is often manifested in the context of endometriosis as ongoing apprehension related to pain flares, upcoming medical appointments, surgical outcomes, fertility concerns, and the impact of the disease on social functioning and daily life (61). The prevalence of clinically significant anxiety in patients with endometriosis-associated chronic pain is substantial, with studies reporting rates as high as 51.0% (26). Consistent with this, a large cohort study found that women with endometriosis were 1.38 times more likely to develop clinically recognized anxiety compared with those without the disease, with this risk being particularly pronounced in younger women (52).

The chronic nature of endometriosis-related pain and the often prolonged diagnostic uncertainty can potentiate anxiety symptoms (62). The presence of anxiety can heighten emotional and cognitive processing of pain, thereby reducing tolerance for discomfort and increasing responsiveness to bodily sensations, establishing a self-reinforcing cycle of pain and anxiety (63). In a cohort of women with endometriosis, 15.8% presented with comorbid anxiety and/or depression prior to surgery; these conditions were consistently associated with severe pain as a surgical indication and greater preoperative healthcare utilization. Increased medication use and frequent physician visits for pelvic pain, including NSAIDs and opioids, were also associated with GAD. Importantly, a retrospective analysis of surgical cohorts further demonstrated that these women have an increased risk of endometriosis-related reoperations, highlighting the significant clinical impact of psychiatric comorbidities on treatment outcomes (43).

Effective management of suspected GAD in women with endometriosis is available and should be delivered by a specialist, typically involving pharmacological treatment with antidepressants and/or anxiolytics combined with evidence-based psychotherapeutic interventions (64). Novel and integrative approaches may be considered in selected cases, under specialist supervision (65). Such strategies may help alleviate endometriosis-associated pain symptoms and consequently improve overall quality of life (63,66).

4.3. Infertility role in depression and anxiety

Infertility is defined as the failure to establish a clinical pregnancy after 12 months of regular unprotected sexual intercourse (67). It represents a major concern for many women with endometriosis, acting as an additional and cumulative source of stress, grief, anxiety, and depressive symptoms (49,68). The complexity of clinical decision-making in infertility cases, which introduces time pressure, repeated procedures, and uncertainty, can amplify distress and conflict in shared decision-making (69,70).

A recent comparative study demonstrated that women with both infertility and endometriosis exhibited significantly higher levels of depressive symptoms and anxiety compared with those experiencing infertility alone; the dual-diagnosis group consistently reported lower scores across all domains of quality of life (QoL) (71). These findings suggest that the psychological burden is not attributable solely to infertility, but rather to the cumulative complexity of the disease trajectory, including the diagnostic process of endometriosis itself, which is often prolonged, stressful, and disruptive. Women are confronted with a chronic

condition lacking a definitive cure and frequently associated with infertility, amplifying emotional distress (13).

Further supporting this perspective, a significantly higher prevalence of depression has been reported among women with endometriosis undergoing IVF or ICSI, suggesting that the type and intensity of assisted reproductive technology may modulate psychological vulnerability (50). Increased depression and anxiety is also experienced after treatment failure, underscoring the importance of considering counseling or psychological support following such outcomes (72). The use of fertility-specific quality-of-life instruments, such as the FertiQoL, is essential for establishing baseline QoL, identifying psychologically at-risk patients, and evaluating the effectiveness of mental health interventions, which have demonstrated beneficial effects on emotional well-being and, in recent systematic reviews and meta-analyses, improved pregnancy outcomes (73,74).

Understanding the role of infertility in the mental health of women with endometriosis therefore requires a holistic and patient-centered approach, encompassing diagnosis, treatment, and the broader life implications of living with a chronic gynecological condition.

4.4. Body dysmorphic disorder and body image dissatisfaction

Body dysmorphic disorder (BDD) is characterized by an intrusive preoccupation with one or more perceived defects in physical appearance, which are often unnoticeable or appear minor to others (59). This intense focus drives repetitive behaviors or mental actions, such as compulsive mirror checking, comparative analysis with others' appearances, and camouflaging perceived flaws, all aimed at alleviating profound distress (59). A formal diagnosis requires that this preoccupation causes clinically significant distress or functional impairment (75). More prevalent in women, BDD is associated with a substantial risk of suicidality (75–77) and high rates of psychiatric comorbidities, including major depressive disorder, social anxiety disorder, and obsessive-compulsive disorder (OCD) (78), frequently co-occurring with chronic pain conditions and emotional distress.

In the context of endometriosis, patients frequently experience a range of physical changes that profoundly disrupt body image and self-perception (79,80). The persistent experience of chronic pelvic pain, combined with the frequent need for invasive medical interventions, can foster feelings of alienation and dissociation from one's own body, paralleling mechanisms observed in anxiety and depressive disorders related to chronic pain, thereby exacerbating underlying psychological vulnerabilities (81–83). This devaluation often extends beyond appearance to bodily function, as individuals with endometriosis report significantly lower appreciation of their body's functional capabilities compared with healthy controls; this finding aligns with qualitative research showing that endometriosis-related pain, heavy bleeding, and fatigue markedly limit engagement in daily activities (83,84), contributing to reduced self-efficacy and emotional well-being.

Visible manifestations, such as chronic abdominal bloating and surgical scars resulting from repeated laparoscopic or laparotomic procedures, further contribute to altered body perception and dissatisfaction (81–83,85). Hormonal treatments for endometriosis may also affect body image, as some women experience side effects such as weight gain and reduced libido (86,87), which can negatively influence self-perception and intensify concerns related to dysmorphic appearance (81), particularly in individuals with pre-existing anxiety or depressive

symptoms. These combined effects on appearance and bodily function mirror the complex body image challenges observed in other chronic pain conditions, including rheumatoid arthritis and inflammatory bowel disease (88,89), where pain severity, fatigue, and functional impairment similarly interact with mood and anxiety symptoms to undermine self-image and self-worth.

Treatment approaches include cognitive behavioral therapy and pharmacological interventions. Given the overlap between body image disturbance, chronic pain, anxiety, and depression, when psychological distress or body image disturbances are suspected in women with endometriosis, referral to specialized mental health care is essential (90).

4.5. Post-traumatic stress disorder and associated traumas

Post-traumatic stress disorder (PTSD) is a persistent and highly debilitating condition that can develop following exposure to traumatic events (91). PTSD is characterized by symptoms such as flashbacks, hypervigilance, avoidance of trauma-related cues, and negative alterations in cognition and mood (59), frequently coexisting with anxiety and depressive disorders. Although the global prevalence of PTSD is estimated at 3.9%, a substantially higher proportion, ranging from 10% to 57%, of individuals with chronic pain meet diagnostic criteria for PTSD (91,92). Relevant exposures in this population include childhood trauma, sexual trauma, invalidation (92,93). This comorbidity can amplify pain severity and emotional distress, reinforcing central sensitization and maladaptive emotional processing, and posing significant challenges for pain management (92).

A key psychological factor contributing to both chronic pain and PTSD is catastrophizing. Catastrophizing represents a critical vulnerability factor influencing the transition from acute to chronic pain and the persistence of PTSD symptoms. This maladaptive cognitive pattern is associated with reduced perceived pain control and heightened emotional distress, overlapping with cognitive processes observed in depression and generalized anxiety disorder, particularly in individuals with comorbid chronic pain and PTSD (94,95).

The clinical trajectory of endometriosis is frequently marked by traumatic experiences, which may culminate in PTSD in a substantial subset of patients. While a case-control study analyzing public health-insurance showed an association between PTSD and increased risk of endometriosis and dyspareunia (96), it is also evident that early-life trauma, including exposure to violence, abuse, or neglect during childhood or adolescence, has been associated with increased risk of endometriosis, PTSD, and other mental health disorders (9,54,93,97,98). Additionally, endometriosis is an estrogen-dependent condition, linking hormonal regulation to stress responsivity; estrogen modulates stress reactivity, and PTSD symptoms in women have been shown to fluctuate across the menstrual cycle in parallel with ovarian hormone variations. These neuroendocrine interactions are consistent with mechanisms implicated in anxiety and depressive disorders in endometriosis. Both conditions are also associated with increased systemic inflammation, suggesting a common biological pathway contributing to symptom persistence and severity (54,99).

Medical trauma represents an additional and often underrecognized contributor to PTSD. Women with endometriosis frequently undergo repeated invasive procedures and may experience medical gaslighting, pain dismissal, and prolonged diagnostic delays, sometimes lasting several years (100–102), which may exacerbate anxiety, depressive symptoms, and

pain-related fear. Reproductive trauma, including infertility and pregnancy loss, commonly associated with endometriosis, can exert a profound and lasting psychological impact (49,103).

Given the substantial psychological burden associated with endometriosis, particularly when PTSD is suspected, timely referral for evaluation and treatment by a mental health specialist is essential. A trauma-informed approach to care is critical, as unaddressed traumatic experiences may render routine clinical encounters distressing or re-traumatizing. Early psychological intervention can help alleviate trauma-related symptoms, reduce emotional distress, attenuate pain amplification, and improve overall quality of life (97,104).

5. Fear-based misinformation, health anxiety, and nocebo mechanisms

The complex nature of endometriosis has given rise to what can be described as a “market of fear,” which operates within and amplifies the bidirectional relationship between chronic pain and mental health vulnerability. This phenomenon exploits the vulnerability of women living with this chronic and debilitating condition. Social media platforms, in particular, have become fertile ground for the dissemination of information, including fear-driven narratives, misinformation, and purported cures lacking scientific validation (105,106). These platforms can perpetuate feelings of anxiety, hopelessness, and sadness among patients (105,107–109), reinforcing pain-related distress and maladaptive emotional processing, and creating an environment of desperation that drives women away from evidence-based treatments toward unproven and often costly interventions (106).

The propagation of fear undermines patient autonomy and complicates clinical decision-making, as a substantial proportion of individuals may disregard professional medical advice when it conflicts with online content (110). This process fuels a feedback loop in which psychological distress intensifies pain perception, while heightened pain further increases emotional vulnerability. This vulnerability is heightened among patients with more severe endometriosis symptoms, both physically and psychologically, who are more likely to engage extensively with social media and therefore face greater exposure to misinformation (111). The dissemination of sensationalized, outdated, or inaccurate information contributes to an environment of negative expectation (112), as reported in recent systematic reviews, which can be understood through the nocebo effect, whereby negative expectations induced by communication exacerbate symptoms, intensify pain perception, generate emotional distress, and promote a state of chronic hypervigilance (113), thereby sustaining both pain and psychiatric symptoms.

6. Diagnostic failure

Diagnostic delay has been identified as one of the most significant systemic failures in the delivery of care for women with endometriosis. The average time to diagnosis ranges from 7 to 10 years globally, reflecting a multifactorial and deeply entrenched problem (25,114,115). Contributing factors include the normalization of menstrual pain, limited clinician awareness, societal stigma, and continued reliance on invasive diagnostic procedures such as laparoscopy (116,117), all of which delay effective pain management and increase psychological burden.

The absence of reliable non-invasive biomarkers and the limited availability of specialized care represent major obstacles to timely diagnosis (25,118). In addition, false-

negative imaging results, particularly in ultrasonography, where peritoneal lesions may go undetected, can lead to premature diagnostic dismissal (6). Such misinterpretations not only delay appropriate treatment but also distort prevalence estimates, thereby influencing diagnostic urgency, health policy priorities, and resource allocation. These delays prolong untreated pain and facilitate the transition from acute to chronic pain states, while simultaneously intensifying psychological distress (117).

Delayed or failed diagnosis constitutes not merely a procedural shortcoming but a psychological burden that directly interacts with pain chronification. As demonstrated by De Corte et al. (114), diagnostic delay is strongly associated with feelings of dismissal and disbelief. Emotional invalidation has been shown to act as a potent catalyst for psychiatric comorbidity, contributing to depression, anxiety, post-traumatic stress disorder, and other mental health conditions (119), which in turn amplify pain perception and reduce coping capacity. This dynamic is often described as medical gaslighting, whereby patients' lived experiences are minimized or discredited by authority figures, frequently through statements such as "it's all in your head" or "it's just period pain" (120,121).

Clinical invalidation erodes trust in healthcare professionals and in patients' own bodily perceptions, often resulting in long-term psychological harm, health anxiety, and trauma. Shapiro and Hayburn (121) further argue that medical trauma, particularly in chronic conditions such as endometriosis, may be more enduring than non-medical trauma, given patients' repeated exposure to clinical environments that replicate invalidation. Within the bidirectional pain–psychiatry framework, this trauma acts as a driver of both emotional dysregulation and pain sensitization. The cumulative consequence of diagnostic failure is a mental health crisis of invalidation, in which delayed diagnosis directly precipitates or exacerbates depression, anxiety, and other psychiatric disorders (122–124). Feelings of shame, frustration, isolation, and pain catastrophizing reinforce a vicious psychological cycle (122,123), maintaining central sensitization and maladaptive threat processing. As patients invest prolonged periods seeking explanations and validation, loss of control and self-esteem intensifies, leading to further deterioration in quality of life.

Systemic inefficiency also imposes a substantial economic burden. Repeated consultations and diagnostic procedures are frequently undertaken for misattributed conditions, including irritable bowel syndrome, urinary tract infections, interstitial cystitis, and psychiatric disorders (25,118). These processes consume significant healthcare resources while failing to provide resolution. Persistent pain combined with unrecognized psychiatric comorbidity further increases healthcare utilization, while time constraints during clinical encounters limit comprehensive symptom assessment, perpetuating cycles of misdiagnosis and patient frustration (125,126).

From both economic and ethical perspectives, these failures underscore the urgent need for diagnostic innovation. The development and implementation of non-invasive, accessible biomarkers (127,128), combined with improved algorithm-based screening tools, hold substantial potential to reduce diagnostic latency (118,129). Earlier and accurate diagnosis may interrupt the bidirectional reinforcement between pain and psychiatric symptoms, alleviating mental health morbidity, reducing progression to chronic pain syndromes, limiting substance misuse, and potentially preventing irreversible disease advancement.

Addressing diagnostic delays is critical to enhancing patient care, managing comorbidities, and alleviating the mental distress experienced by patients. Implementing a

multifaceted intervention strategy could be beneficial in addressing these concerns (130). Providing training to physicians, particularly those from other disciplines, along with mandatory continuing education, could increase awareness among physicians. Improving and adapting endometriosis guidelines would help simplify diagnosis. Furthermore, qualitative research in primary care advocates that implementing a diagnostic decision tool could especially help non-gynecologists make diagnoses or refer patients for further evaluation (130). This recommendation includes implementing evidence-based approaches to endometriosis to combat misinformation.

7. Prognosis and recurrence

Endometriosis frequently recurs or progresses. Recurrence rates include 15% after surgical resection of peritoneal lesions or colorectal surgery for deep endometriosis (131,132) and 6–67% after medical therapy (133). Treatment is further complicated by approximately 30% of patients being unresponsive to progestins (134,135). A meta-analysis revealed that 5–59% of women continue to experience pain after treatment, and 11–19% can report no relief (135).

The association between stress and increased pain severity has been well-documented in the scientific literature. Higher stress levels have been shown to be associated with more severe pain in patients with endometriosis (136). This persistent anxiety regarding disease progression or recurrence can lead to adverse physical and mental health outcomes (137,138). The proliferation of fear-based misinformation in the digital ecosystem further intensifies these risks by promoting narratives of inevitable recurrence and limited treatment options. In this context of persistent pain and mental health burden, novel therapeutic modalities are needed for managing endometriosis-associated pain and its mental health comorbidities (65,139–141).

8. A call for integrated care and governmental policy

Effective management of endometriosis necessitates a truly multidisciplinary approach and coordinated governmental action grounded in a biopsychosocial framework. This integrated strategy must systematically incorporate mental health professionals, including psychiatrists and psychologists, to address the significant psychiatric comorbidities associated with the disease and their bidirectional interaction with chronic pain and functional impairment. Governments play a vital role not only by funding and standardizing specialized multidisciplinary endometriosis centers, but also by leading comprehensive public and professional education initiatives focused on endometriosis and its psychiatric comorbidities. Such actions can reduce stigma, promote early recognition of both physical and mental health symptoms, and facilitate timely, integrated intervention, ensuring patients receive comprehensive, evidence-based, and ethically grounded care.

Future research should prioritize the development and validation of screening instruments tailored to identify psychiatric comorbidities within endometriosis populations, particularly those linked to pain severity and disease burden. Policy-oriented research is essential to evaluate the effectiveness of government-sponsored integrated care models, examine their impact on pain outcomes, mental health trajectories, and healthcare utilization, and assess the role of public awareness campaigns in reducing stigma and improving access to

mental health services. Additionally, training programs for physicians and multidisciplinary teams should be evaluated with respect to their ability to recognize and manage the bidirectional reinforcement between pain and psychiatric symptoms. The advancement of noninvasive and more accurate diagnostic tools, including the application of artificial intelligence, remains crucial to reducing diagnostic delays and thereby mitigating downstream mental health consequences. Novel therapeutic approaches that address both pain and psychiatric vulnerability are also needed. Together, these priorities constitute a call for systemic recognition of endometriosis as a complex, multisystem condition and for integrated reforms that address all facets of the disease, from diagnosis to multidisciplinary treatment and sustained psychosocial support.

In order to translate the conceptual insights offered in this review into tangible improvements in care, we propose an implementation framework for integrated endometriosis services. At a minimum, such a service should include: gynecological care with specialized expertise in endometriosis management to facilitate accurate diagnosis and personalized treatment planning; comprehensive pain management; pelvic floor physiotherapy to assess and treat musculoskeletal contributors to persistent pain; mental health support, including access to psychology and psychiatry professionals who can screen for and manage anxiety, depression, trauma-related symptoms and pain-related distress; fertility evaluation and support to address reproductive goals and related psychosocial stressors. Incorporating these core components into a coordinated care pathway could reduce diagnostic delays and symptom burden, and improve psychological outcomes and health-related quality of life.

9. Conclusion

Endometriosis imposes a substantial mental health burden, with affected women facing a heightened risk of depression, anxiety, body dysmorphic disorder, and post-traumatic stress disorder. These vulnerabilities are not isolated phenomena but are dynamically intertwined with chronic pain through bidirectional neurobiological and psychosocial mechanisms. Systemic failures in care delivery, including diagnostic delay and medical invalidation, further exacerbate this vulnerability, while the emergence of a “market of fear” amplifies distress and undermines evidence-based decision-making.

Public health policy implementation is therefore essential to support integrated, multidisciplinary models of care that address both pain and mental health, strengthen mental health services within endometriosis care pathways, and actively counter misinformation and fear-driven narratives. Only through coordinated systemic reform can the cycle of pain, psychological suffering, and invalidation be interrupted, allowing patients to receive the comprehensive, ethical, and person-centered care they require.

Competing Interest: JAC is coinventor of the patent “Fluorinated CBD compounds, compositions and uses thereof. Pub. No.: WO/2014/108899. International Application No.: PCT/IL2014/050023,” Def. US number Reg. 62,193,296; July 29, 2015; INPI on August 19, 2015 (BR1120150164927; Mechoulam R, Zuardi AW, Kapczinski F, Hallak JEC, Guimaraes FS, Crippa JAS, Breuer A). Universidade de São Paulo (USP) has licensed this patent to Phytects Pharm (USP Resolution No.15.1.130002.1.1) and has an agreement with Prati-Donaduzzi to “develop a pharmaceutical product containing synthetic CBD and prove its safety and therapeutic efficacy in the treatment of epilepsy, schizophrenia, Parkinson’s disease, and anxiety disorders.” JAC is coinventor of the patent “Cannabinoid-containing oral pharmaceutical composition, method for preparing and using same,” INPI on September 16, 2016 (BR 112018005423-2). JAC has received personal consultation fees from EaseLabs, BSPG-Pharm, and PurMed Global in the past. JAC received speaking fees from Torrent, Aché, Green Care Store, Janssen and consultation and speaking fees from EaseLabs. MSA is on the advisory board for Roche, Forasevit and is consultant Chugai Pharmaceuticals. All the other authors have no relevant financial or non-financial interests to disclose.

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Figure Legend:

Figure 1. Biopsychosocial model of mental health outcomes

